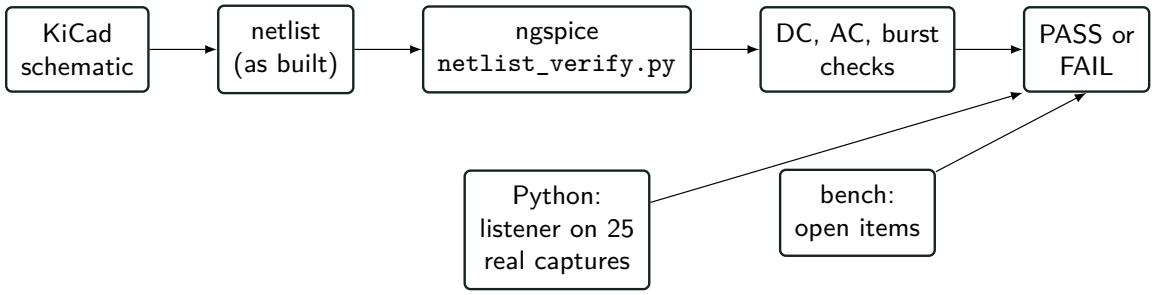
Slide 13: verification and next steps

What the simulation proves and what is still open

ADS7057 test board v3 — study notes

1 How it is checked



`sim/netlist_verify.py` reads the KiCad netlist of the board as built, so a wrong net or value shows up. It builds an ngspice circuit with vendor models and runs a DC operating point, an AC sweep and a burst test. A second script runs the listener thresholds on 25 real captures.

2 Results

What is checked and what is open

DC operating point, all rails (18 checks)	PASS	ngspice, as-built netlist
Gain 12.3 dB at 250 kHz; −40.5 dB at 10 kHz; −24 dB at 2 MHz	PASS	ngspice
Listener thresholds: WAKE_N 35 mV, BIG_N 111 mV at SIG_IN	PASS	ngspice, 5 burst levels
ERC, DRC, schematic against board	PASS	KiCad, 0 errors
Head and bias-T at 3.3 V; listener on 25 real captures	SIMULATED	ngspice, Python
ZedBoard 3.3 V rail noise (TP11, TP12)	OPEN	measure
PMOD fit on adjacent ports	OPEN	print 1:1, test
THS4531 stable with 400 pF + 20 Ω	OPEN	LTspice / bench
First board on the bench	OPEN	manual steps 49 to 52

Check	Result	Limit
All 18 DC nodes (rails, mid-rails, ladder)	within limits	±1 to 3 %
Gain SENSOR IN to ADC at 250 kHz	12.28 dB	11.5 to 13
Loss at 10 kHz and 2 MHz	40.5 dB, 24.0 dB	38 to 50, 18 to 40
Current from the PMOD	24.8 mA	15 to 40
Burst, 45 mV generator	WAKE_N low, BIG_N high	as designed
Burst, 112 mV generator	both low	as designed

Result: all pass. The simulation proves that the netlist does what the design intends. It does not prove the real parts, the layout or the sensor spectrum.

3 Open items, in order of risk

1. **THS4531 with** 400 pF **and** 20 Ω**.** The datasheet expects about 2 pF between the outputs. Stability with this load is not confirmed yet. Run the LTspice netlist, then look at an output step on the bench.
2. **Anti-aliasing margin.** Only 8 dB at 800 kHz (slide 6). Measure the real sensor spectrum with the new head.
3. **Listener margin.** Noise peak 26 mV against 35 mV. Set the thresholds from bench data.
4. **ZedBoard 3.3 V rail noise** (TP11, TP12) and the **PMOD fit** on adjacent ports (print 1:1).
5. **ADC decoupling:** the 100 nF DVDD capacitor is 4.5 mm from the pin.
6. **Manual fixes:** the head test text says VREF settles in 0.2 s; with a 1 MΩ divider it takes about 25 s. The driver input corner is 119 Hz, not 159 Hz (slide 7).

4 Next steps

1. Bring-up in this order: rails (manual step 49), a 250 kHz burst at J1 (step 50), the response sweep (51), the listener (52).
2. Scale to 16 channels with the backplane (FMC adapter and two 8-slot modules, in `hw/backplane/`). Open items there: JLC placement of the FMC plug, card guides, SCLK integrity.
3. FPGA side: a scheduler for the two modes (continuous sampling of a few plants, wake-up of all 16).